PostgreSQL 16 Advanced Indexing & EXPLAIN Plans • STUDENT EXAM COPY

Course: PostgreSQL 16.6 • College: DPU • Student Name: _____

Q1. What does the PostgreSQL 'EXPLAIN' statement output for a given query?

A) The execution plan chosen by the query planner, showing sequential or index scans

B) The complete table schema in JSON format

C) A list of all connected client IP addresses

D) The estimated hardware cost in USD to run the server

Q2. Why is deterministic output normalization critical in automated code verification for PostgreSQL?

A) It masks memory addresses, timestamps, and thread IDs so identical logic yields identical results

B) It rewrites slow algorithms into faster ones automatically

C) It disables unit testing to make test suites run faster

D) It converts console output into binary machine code

Q3. What is the primary objective of mutation testing in exercise validation?

A) Proving that student-facing test suites successfully fail when code logic is intentionally mutated/broken

B) Randomly deleting student files to test backup recovery

C) Speeding up Docker container build times

D) Encrypting source code with AES-256

Q4. In PostgreSQL, what is the main consequence of executing untrusted code without container sandbox isolation?

A) Direct host filesystem compromise, resource exhaustion, and external network leaks

B) Syntax highlighting colors changing randomly

C) The compiler running 100x faster than expected

D) PowerPoint slide fonts changing to Comic Sans

Q5. When configuring container limits for isolated verification, why is '--network none' strictly enforced?

A) To prevent code from downloading malicious payloads, leaking credentials, or contacting external servers

B) Because Docker containers cannot support networking on Linux

C) To decrease the size of Docker image layers on disk

D) To disable JSON serialization in the API

Q6. What is the recommended approach for handling deprecation warnings in current production PostgreSQL?

A) Treat warnings as errors in CI/CD pipelines and migrate to the approved modern replacements

B) Pin older runtime versions indefinitely to avoid updating

C) Turn off compiler error logging across the entire company

D) Use reflection to bypass deprecated method signatures

Q7. Which practice is recommended when engineering scalable PostgreSQL applications for PostgreSQL 16 Advanced Indexing & EXPLAIN Plans?

A) Use verified idioms, explicit types, and automated unit test assertions

B) Suppress all compiler and deprecation warnings to simplify logs

C) Rely entirely on global mutable state to avoid passing parameters

D) Execute all database calls synchronously inside UI or main thread loops

Q8. Which practice is recommended when engineering scalable PostgreSQL applications for PostgreSQL 16 Advanced Indexing & EXPLAIN Plans?

A) Use verified idioms, explicit types, and automated unit test assertions

B) Suppress all compiler and deprecation warnings to simplify logs

C) Rely entirely on global mutable state to avoid passing parameters

D) Execute all database calls synchronously inside UI or main thread loops

Q9. Which practice is recommended when engineering scalable PostgreSQL applications for PostgreSQL 16 Advanced Indexing & EXPLAIN Plans?

A) Use verified idioms, explicit types, and automated unit test assertions

B) Suppress all compiler and deprecation warnings to simplify logs

C) Rely entirely on global mutable state to avoid passing parameters

D) Execute all database calls synchronously inside UI or main thread loops

Q10. Which practice is recommended when engineering scalable PostgreSQL applications for PostgreSQL 16 Advanced Indexing & EXPLAIN Plans?

A) Use verified idioms, explicit types, and automated unit test assertions

B) Suppress all compiler and deprecation warnings to simplify logs

C) Rely entirely on global mutable state to avoid passing parameters

D) Execute all database calls synchronously inside UI or main thread loops

Q11. What is the exact output produced by executing this PostgreSQL code?

```
SELECT 10 + 25 AS total;
```

A) Compilation Error / Syntax Warning

B) total: 35

C) null

D) Runtime Exception: IndexOutOfBounds

Q12. What is the exact output produced by executing this PostgreSQL code?

```
SELECT UPPER('verified') || '_2026' AS code;
```

A) Compilation Error / Syntax Warning

B) code: VERIFIED_2026

C) null

D) Runtime Exception: IndexOutOfBounds